National

Operational Excellence Standard (OES) — Premise wiring installation

HOW22915 | MODIFIED 11/28/2025



Summary

This standard will review all requirements for the correct installation at the customer premise from the Service Distribution Point to the outlet/customer premise equipment delivering services.

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What's changed in this article

Date	Summary of change
September 11, 2025	Added to Celestial.
October 16, 2024	Updated minimum bending guidelines.
March 21, 2024	Conducted scheduled review and updated: • Added vinyl siding clip CIFAs. • Updated to Ethernet wiring. • Added opening sentences to At a glance section.
March 8, 2022	Conducted scheduled review and updated.
November 17, 2020	Reviewed for content needed to include commercial property.
February 21, 2020	Released document.

At a glance

This article contains the premise wiring installation OES.

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Required tools and hardware

Description	CIFA number
CAT 5 Lock-In Vinyl Siding Clip Horizontal - Black	25658
CAT 5 Lock-In Vinyl Siding Clip Horizontal - White	25659
RG-6 Lock-In Vinyl Siding Clip Horizontal - Black	19466

Description	CIFA number
RG-6 Lock-In Vinyl Siding Clip Horizontal - White	19467
RG-6 Lock-In Vinyl Siding Clip Vertical - Black	24588
RG-6 Lock-In Vinyl Siding Clip Vertical – White	24589

Guidelines

- This article does not substitute for any local, state, or federal code or requirements, but should be used as a supplement to these.
- **Deviations:** Practices and configurations outlined in this document are the Comcast standard and may not be modified in their implementation. Any deviations from the standards prescribed in this document shall be approved by Comcast Corporate Technical Operations.
- **Safety:** All personnel are required to always comply with Comcast's security and safety regulations.
- **Workmanship:** All work performed will be of the highest possible workmanship. All work will adhere to Comcast's standards, local, state, and federal codes and requirements and be fully compliant to any stipulations made by the landlord or agent in control of the property.
- Within this section the terms "cable, cables or drop" may refer to either a coaxial, twisted pair, ethernet, or fiber component. When required, the use of each term will be defined based in the specification.

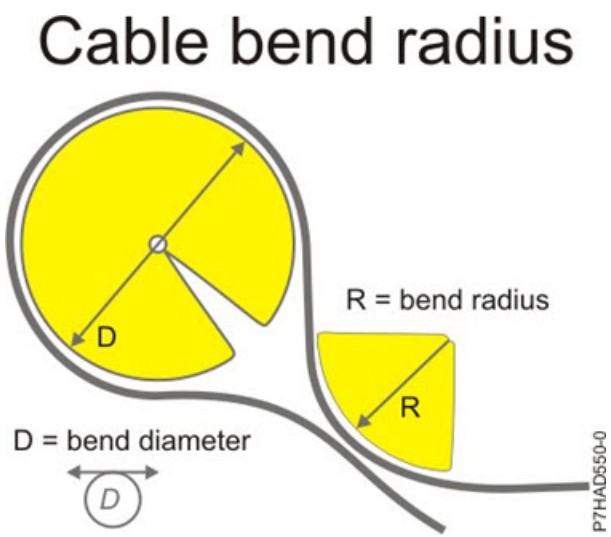
- The service distribution point shall establish the demarcation between the outside drop network and the internal premise distribution wiring.
  - See OES service distribution point for additional information.
- Coaxial cable installed after the service distribution enclosure or after the point of bonding at the bond block must use type indoor cable except where a higher-rated cable, riser or plenum, is required.
- All twisted pair and Ethernet wiring must be rated for indoor and outdoor use.
- All coax cable shall meet all the listing requirements of NEC Article 820. This classification will be visibly marked on the outside of the cable.
- All fiber optic cable shall meet all the listing requirements of NEC Article 770.
- All other communication cables shall meet all the listing requirements of NEC Article 800.
- See OES approved cable connectors and hardware for additional guidance.
- Type plenum cable and/or micro duct must be used in environmental air handling plenums and ducts and shall meet all the listing requirements of the NEC Article for that cable type.
- Messengered or PE jacketed underground flooded cable must never be used inside the customer premise.
- Cables not approved or not certified by Comcast Engineering for outdoor use must never be used for exterior wiring as they are not UV/ozone-stable and will deteriorate when installed outdoors.

Outdoor cable routing and attachment standards and specifications

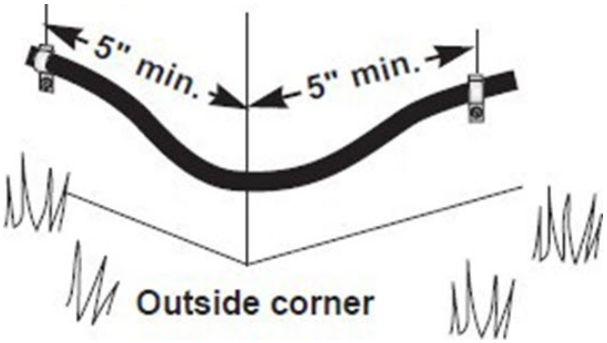
- Cable routing must follow the natural lines of the structure and must be done to minimize the visibility of the cable.
- Cable, drop splitters, premise boxes, or other cable hardware must **never** be attached to water pipes, gas lines, electrical power lines, electrical power masts, electrical power conduit, or any other object to which its attachment might interfere with other services or utilities.

- Cables, drop splitters, or other cable hardware must never be attached to rain gutters or downspouts.
- Cable must be routed and fastened to maintain the minimum bend radius of the cable at all times.

Cable type	Minimum bend diameter	Minimum bend radius
RG - 6	5 inches	2 ½ inches
RG - 11	9 inches	4 ½ inches
CAT6	2 inches	1 inch
Ballistic fiber	2 inches	1 inch

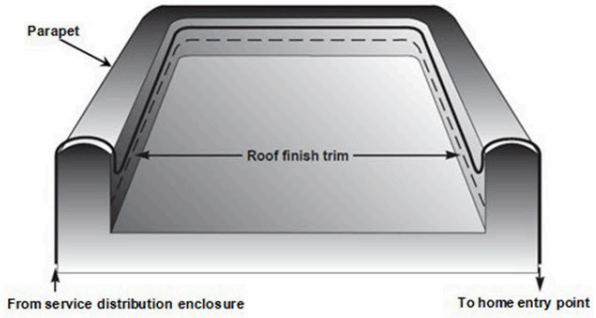


- Cable must be fastened or clipped every 18 to 24 inches if routed horizontally and fastened or clipped every 30 to 36 inches if routed vertically.
  - When routing cables on aluminum or vinyl siding, the appropriate vertical or horizontal siding clips must always be used as to not make holes in the siding.
  - Cable shall not be tucked into any corner channel or j channel of vinyl siding.
  - Vinyl siding needs room to expand and contract with changes in temperature. Not following the above guidelines may result in warping or damage of the siding.
- **Never** make a 90-degree square corner with the cable to avoid damage to the cable.



- When routing cable from the outside service distribution point or enclosure to the premise entry point for premises built with a flat roof, do not lay cable on the roof.
- The only exception will be if there is a barrier extending above the roof surface forming a parapet. If a parapet is available, Cable must be run along the top inside edge and fastened above the roofing material 3 to 4 inches below the edge of the parapet. Never drill through the roof. The Cable must be then run over the edge of the parapet and down the wall to the premise entry point.
- Local and OSHA safety guidelines for rooftop work must be followed at all times. Only technicians who are qualified through the Comcast Rooftop Training Program are allowed to access rooftops.
- When routing Cable from the outside service distribution point or enclosure to the premise entry point, the cable shall be routed through an entry point for the purpose of cable or communication wire entry. The cable must **never** be routed through any

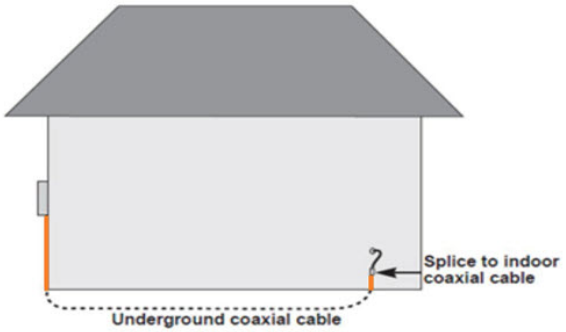
other means such as a window, vent pipe, soffit, vent, gas line, electrical line etc.

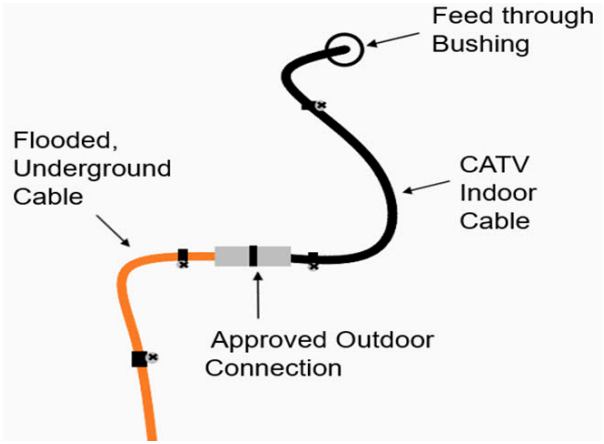


Splicing wiring and cables

Coaxial cable

- All cables between the service distribution point and the entry to the premise (homerun) must be free of splices or splits.
- The only exception is when routing cable underground from the outside service distribution point or enclosure to the premise entry point. Flooded underground cables must be used and connected to inside CATV indoor cables using an approved outdoor connection. **Never** use underground cables into the premise.





Twisted pair and Ethernet

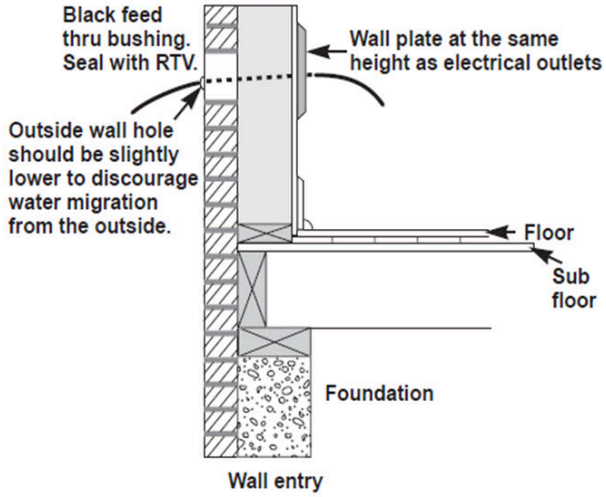
- All new twisted pair wiring shall be free of any splices, except when transitioning to equipment. In these cases, approved connections must be utilized.
- Existing wiring for voice cabling may be spliced utilizing approved splices.
- Existing wiring for ethernet cabling shall be free of any splices except when transitioning to equipment. In these cases, approved connections must be utilized.
- For Xfinity Home use cases see OESx-Xfinity Home installation.

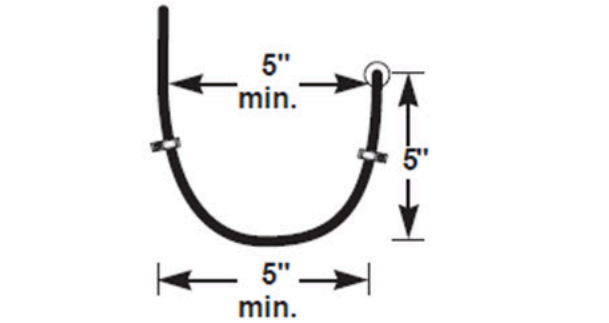
Fiber cable

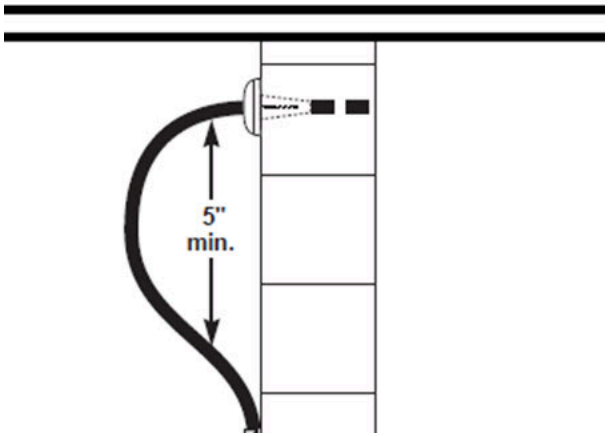
All new fiber cables shall be free of any splices, except when transitioning to equipment. In these cases, approved connections must be utilized.

Premise entry of wiring

- To prevent water damage, all exterior entry holes must be drilled so that the outside of the hole is slightly lower than the hole on the inside of the wall.
- Outside entry hole drilling direction must consider the outside construction material of the premise.
  - Drilling shall be from the inside out for wood or vinyl siding.
  - Drilling from the outside in for all other materials, including metal, slate, stucco, hard asphalt, or brick.
- Always check the inside location for plumbing, electrical wires, gas lines, and any other hazards before drilling.
- Drilling of a foundation wall is acceptable only when all other means for cable entry into the premise are exhausted. Customer and supervisor approval is required in these situations.
- An outside feed-through bushing is required to be installed over the cable and must be sealed with RTV sealant and/or caulked around the inside edge of the feed-through bushing and around the cable going into the bushing. This will prevent water, air, and insects from entering through the outside wall.
- A minimum diameter drip loop, in accordance with the bending radius for the cable being used, must be formed below all outside entry holes anywhere cable enters the premise. Before fastening the cable to the premise wall, and **before** insertion through any outside entry hole. The below example is for series 6 cable.
- Cable **must always** enter the premise from below the level of the outside entry hole, **never** from above the entry hole, to prevent moisture from traveling down the cable and into the premise.





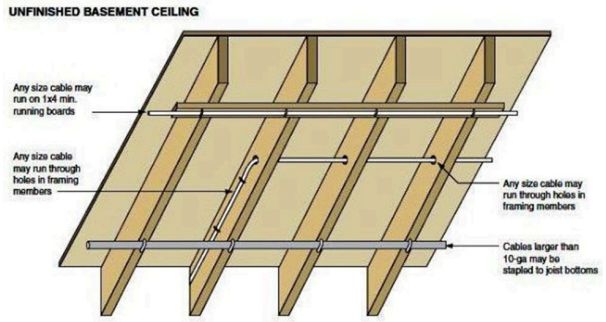


- Entry holes in mobile homes or construction trailers must be through the floor only. Never drill through the outside wall.

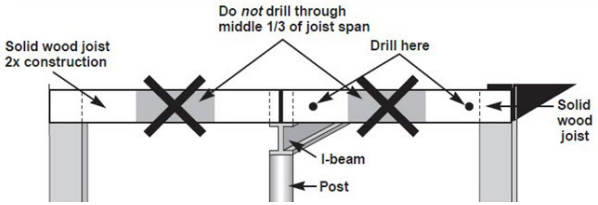
- Always install a wall bushing in a floor entry hole for mobile homes or construction trailers before inserting the cable.

Indoor cable routing and attachment options

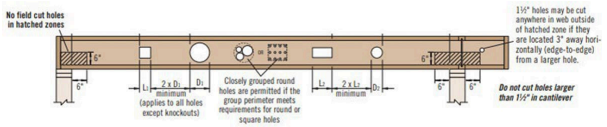
- All new outlets must be home run to the central primary service distribution point.
- Never run cable under carpets, through door frames, through windows or window frames.
- Cable routed through any floor/ceiling joists must NEVER be routed through holes occupied by any electrical wires.
- Cable routes must maintain a minimum distance of two inches from any electrical wiring.



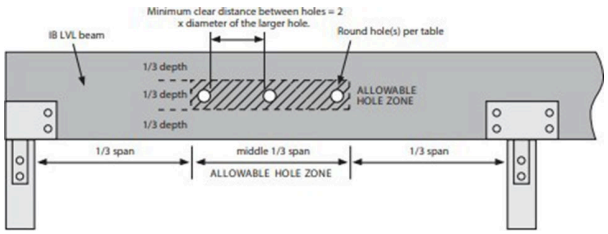
- Cable routed through a basement must be routed parallel or perpendicular to the floor joists, never diagonally. If running cable parallel to the joists, the cable must be attached to the side of the joist.
- Cable routed through a basement perpendicularly to the floor/ceiling joists must be routed through holes at least two inches from the bottom of the joist or on a running board. Cable may never be attached to the bottom of a floor/ceiling joist.
- Drilling through the middle one-third of any floor/ceiling solid wood joist span will weaken the solid wood joist and is **prohibited**.

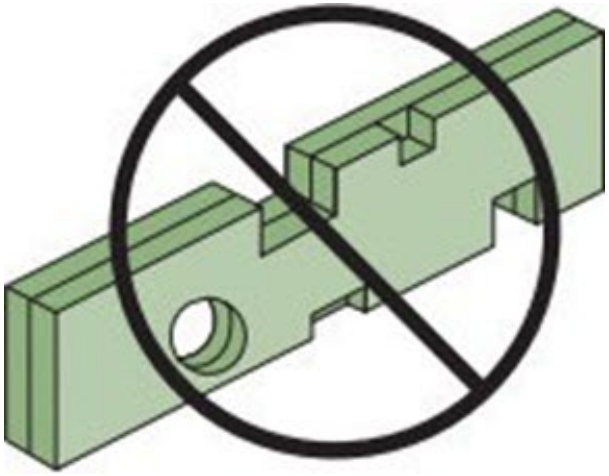


- When routing cable through a basement perpendicularly to TJI joists with perforated knockouts.
  - Preferred method is to use the perforated knockouts in the web of the joist
  - Holes maybe drilled if they are 6 inches away from any support post or foundation. They are less than 1 ½ inches in diameter and located 3 inches away horizontally (edge to edge) from a larger hole.
  - Never cut or notch the upper or lower flange of a TJI joist.
  - Holes less than 1 ½ inches in diameter can be cut in cantilever. However, do not cut holes in any cantilever reinforcement.



- Cable routed through a finished or partially finished basement perpendicularly to laminated veneer lumber joists must only be routed through holes drilled as indicated in the figure below.
  - Hole(s) must be located completely in the allowable hole zone.
  - No more than three holes are allowed per span.



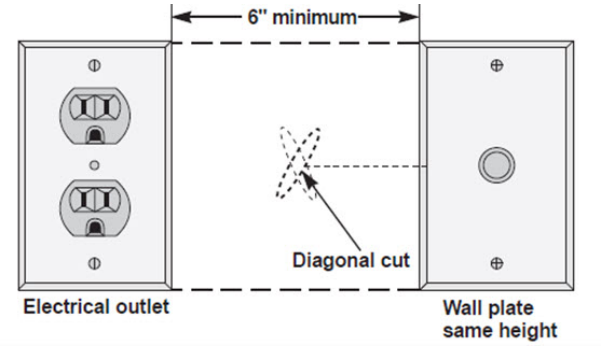


- If a suspended ceiling is in use, cables must not lay directly on the ceiling. All cables must be securely attached above the ceiling or in communication cable trays or raceways.
- Any penetrations through a firewall must be closed with approved fire stopping material.

Termination point installation

- All cabling should terminate to an approved termination point (i.e., wall plate, termination panel, etc.).
- Connections from the termination point to CPE shall be made via an approved jumper.
  - An exception is allowed for mobile homes and construction trailers.
- Distance between the wall plate and the CPE should not be more than 6 feet.
  - If equipment cannot be located within six feet away from the outlet, the jumper cable must be securely fastened to eliminate any safety hazard.
- Wall plates must be level mounted at the same height as the electrical outlets in the room and no closer than six inches from any electrical outlet.
- Wall plate mounting brackets must be used when installing an outlet.

- See your local division engineering for additional guidance or exceptions.





- Coaxial wall plates must come with an F-81 barrel connector or F-81 keystone. The barrel connector must be mounted with the hexagonal portion of the barrel set in the hexagonal recess in the wall plate to prevent the barrel from turning when a connector is installed or removed.
- Twisted pair wall plates must incorporate a Registered Jack (RJ) appropriate for the type of cable or connection needed.
- Fiber wall plate optical jacks must incorporate an auto-shutter or similar stray light protection.

References

- National Electrical Code
  - Article 820
  - Article 770
  - Article 800
- Occupational Safety and Health Administration service distribution point OES
- Approved cable, connector, and hardware OES

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